1.0 PULSE WIDTH MODULATION UNIT (PWMU)

1.1 Features

- Three independent PWM generators
- Two PWM outputs per each PWM channel
- Two current acquisition unit (CAU) trigger signals per each PWM generator
- Independent control of PWM outputs
- Centre and edge aligned modulation mode of operation
- Dead time generation
- Complementary PWM signal Generation
- Settings to invert PWM signal Automatically
- 12.5 ns PWM resolution at 80 MHz peripheral clock
- PWM frequency of 4.9 kHz up to 5 MHz
- 12-bit PWM resolution at 20Khz
- Phase shift in centre-aligned mode
- Compatibility with both single and multiple shunt systems
- Phase disable signal to support block commutation control for high-speed motors

1.2 Introduction

The PWM Module is designed to support the control of various types of motors such BLDC and DC motors. The main functionality of the PWM module is to generate PWM signals to the gate drive unit (GDU) to drive the external MOSFETs. In addition to the PWM signals, there are dedicated current sampling trigger signals generated by this block which is tailored for field-oriented control (FOC) application. These signals support both single and multiple shunt systems. The flexibility in the configuration settings combined with new introduced features makes this module suitable for variety of motor control methods such as field-oriented control(FOC), Hall sensor control and back-emf based sensorless control. The main blocks forming the PWM module is shown in the next section.

1.3 Functional Description

The top-level architecture of the PWM module is shown in Figure 1-1. The clock frequency (f_{PWM}) of this module is 80Mhz. The register settings are set by the AHB bus. The input control setting from the Advanced Motor Control Timer module (AMCT) can be utilized for hall sensor based and BEMF based type of control algorithm. The PWM outputs of this module are connected to general purpose input output (GPIO) driver which connects the PWM signals directly to the gate driver unit (GDU) in order to drive the external MOSFETs. The analog to digital converter normal and calibration trigger signals (ADCANT and ADCACT) for each PWM phase is connected to Current Acquisition Unit(CAU) to sample the current at a desired point with respect to the generated PWM period cycle making it suitable for field oriented control (FOC) application. The pwm generator is capable of generating complementary PWM outputs with two modes of operation , Centre aligned and edge-aligned. There is also a dedicated dead time control unit to ensure no short circuit occurs when the complementary signals Hx and Lx are used to drive the external MOSFETs. There is also a PWM_Centre signal which is set high at the centre of PWM cycle and is set low at the start of PWM cycle. This signal is connected is connected to digital acquisition unit (DAU), advanced motor control timer unit (AMCT) and GPIO. Refer to each respective module for further information.

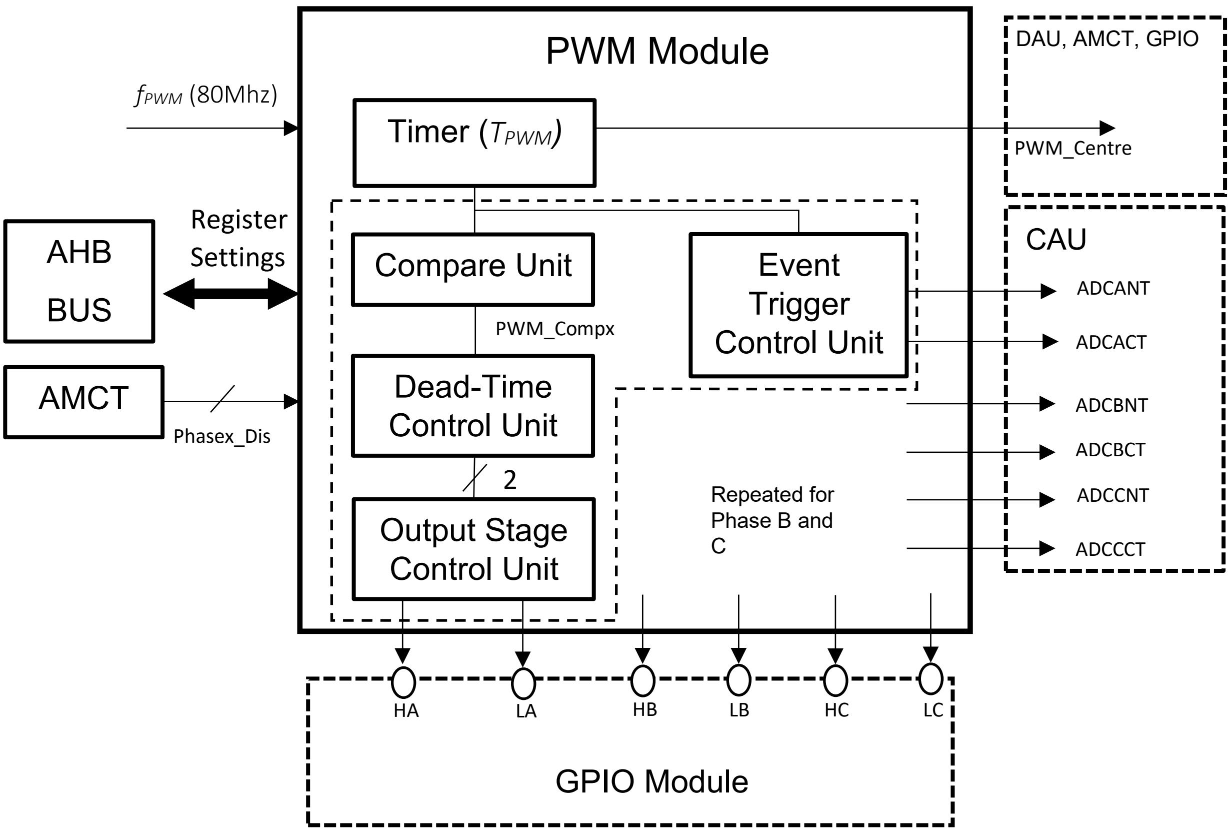

Figure 1-1: Block Diagram of the PWM Module

1.4 Operating Timer (T_{PWM})

The clock source (f_{PWM}) to the operating timer is 80MHz. The timer T_{PWM} contains a 14-bit counter which is used as a reference signal for the compare and event trigger control units. T_{PWM} has two modes of operation: centre aligned, and edge aligned. In both modes, the period of the timer T_{PWM} is defined by the PERIOD parameter in the PWM PERIOD register and can be calculated according to the formula below:

$$T_{PWM}\ (Period) = \frac{PERIOD}{f_{PWM}}\ \mathrm{s}$$

For example if it is desired to generate a signal with a period of 50µs (20KHz) , the PERIOD can be calculated as below assuming f_{PWM} of 80MHz:

$$PERIOD = 50\ \mu s \times 80\ MHz = 4000$$

1.4.1 Centre-aligned Mode Carrier Signal

The timer T_{PWM} is set to centre-aligned mode when PWMMOD is set to ‘1’ in Control1 register. Figure 1-2 shows the functional behaviour of the timer in this mode. TCOUNT counts up for half of the T_{PWM} period and counts down for the other half. In this mode of operation, if PERIOD is set to an odd number, the number will be rounded down to the closest even number. For example, a PERIOD of 301 will be rounded to 300. PWM_Centre is set high at (T_{PWM} (period))/2 and it is set low at the start of the new T_{PWM} period. The Carrier parameter in the PWM Carrier register is incremented on every rising edge of the clock (f_{PWM}) and it resets to 1 when Carrier has reached to the value defined by the PERIOD. For example in Figure 1-2 the PERIOD is set to 16_d, the Carrier signal is incremented at every rising edge of the clock (f_{PWM}) and resets once it reaches to 16_d . To start this mode of operation PERIOD needs to be set to a value greater than 15_d.

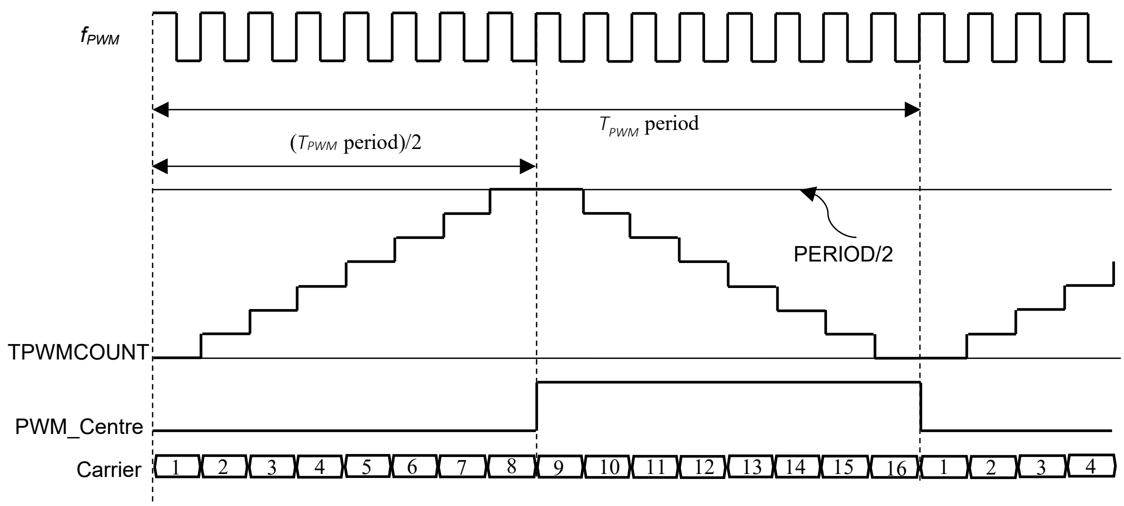

Figure 1-2: Timer Operation in Centre-Aligned Mode

1.4.2 Edge-Aligned Mode Carrier Signal

The timer T_{PWM} is set to edge-aligned mode when PWMMOD is set to 0 in Control1 register. Figure 1-3 shows the functional behaviour of T_{PWM} in edge-aligned mode. In this mode T_{PWM} counts up from zero to PERIOD value. PWM_Centre is set high at (T_{PWM} period)/2 and it is set to low at the start of the new T_{PWM} period. The Carrier parameter in the PWM Carrier register is incremented on the rising edge of clock (f_{PWM}) and it resets to 0 when Carrier has reached to the value defined by the PERIOD. Figure 1-3 shows TPWMCOUNT behaviour in edge-aligned mode. To start this mode of operation PERIOD needs to be set to a value greater than 15_d.

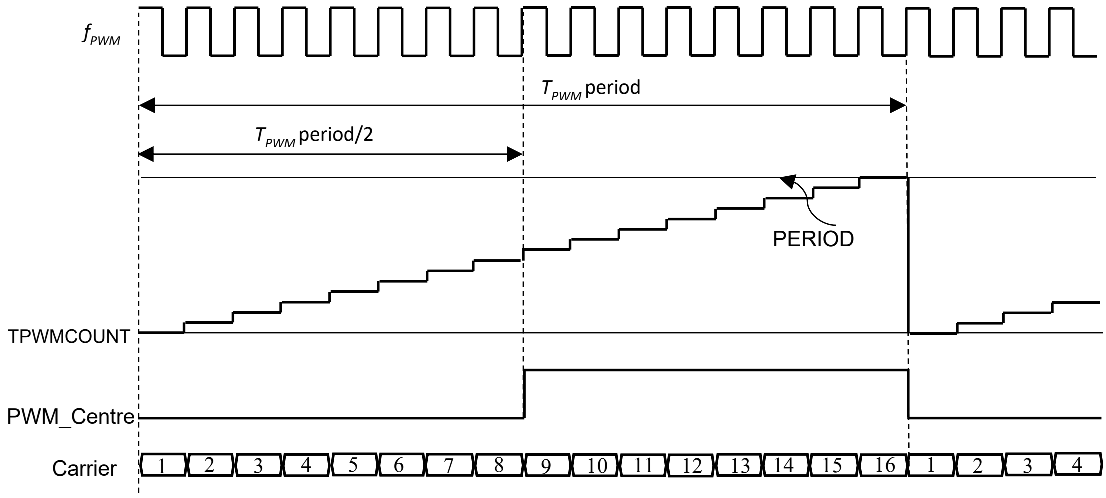

Figure 1-3: Operating Timer in Edge-Aligned Mode

1.5 Compare Unit

The compare unit generates the reference signal to the dead-time control unit to generate the complementry PWM signals to the GPIO unit. The port control is then processed in GPIO unit in order to send the compelmentry signals to the gate drive unit (GDU) to drive the external MOSFETs. The operation of the compare unit depends on the T_{PWM} mode of operation. The following two sections show the functional behaviour of the compare unit in centre and edge aligned mode.

1.5.1 Centre-aligned mode

Figure 1-4 shows the functional behaviour of the compare unit in the centre-aligned mode. The compare unit in centre-aligned mode (PWMMOD=1), compares two variables, DutyCycle1x and DutyCycle2x, to TPWMCOUNT to generate the output PWM_Compx to the dead-time control unit, where 'x' represent one of the three independent PWM generators. DutyCycle1x and Dutycyle2x are programmed by PWM duty-cycle register. DutyCycle1x is compared to the first half of TPWMCOUNT and DutyCycle2x is compared to the second half of the TPWMCOUNT as shown in Figure 1-4. When Phasex_Inv in control1 register is set to 0, PWM_Compx is set high when DutyCycle1x/2 and DutyCycle2x/2 are greater than the TPWMCOUNT and is set low when smaller than or equal to TPWMCOUNT. Inversly, when Phasex_Invert is set to 1 , PWM_Compx is set high when DutyCycle1x/2 and DutyCycle2x/2 are smaller than or equal to the TPWMCOUNT and is set low when greater than the TPWMCOUNT. The duty cycle of the PWM_Compx signal is calculated as below depending on the Phasex_Inv signal.

When Phasex_Inv = 0

$$PWM\_Compx(Duty\ Cycle) = \frac{\frac{\text{DutyCycle1x}}{2} + \frac{\text{DutyCycle1x}}{2}}{PERIOD} \times 100\ \%$$

When Phasex_Inv = 1

$$PWM\_Compx\ (Duty\ Cycle) = \frac{\frac{(PERIOD-\text{DutyCycle1x})}{2} + \frac{(PERIOD-\text{DutyCycle2x})}{2}}{PERIOD} \times 100\ \ \%$$

For example if it is desired to achieve 30% duty cycle for a T_{PWM} period of 50µs,

DutyCycle1x and DutyCycle2x can be calculated as below when Phasex_Inv = 0.

From section 1.4, PERIOD should be 4000 to achieve T_{PWM} period of 50µs.

$$\text{DutyCycle1x} + \text{DutyCycle2x} = \frac{30 \times 2 \times 4000}{100} = 2400$$

In order to preserve symmetry in Figure 1-4, DutyCycle1x = DutyCycle2x.

Therefore :

$$\text{DutyCycle1x} = \text{DutyCycle2x} = 1200$$

Important Notes:

If PERIOD, DutyCycle1x or DutyCycle2x are odd numbers, they are rounded down to a closest even number. If DutyCycle1x or DutyCycle2x are greater than PERIOD, they are respectively clamped to PERIOD.

The operations starts once the PERIOD set to a value greater than 15_d.

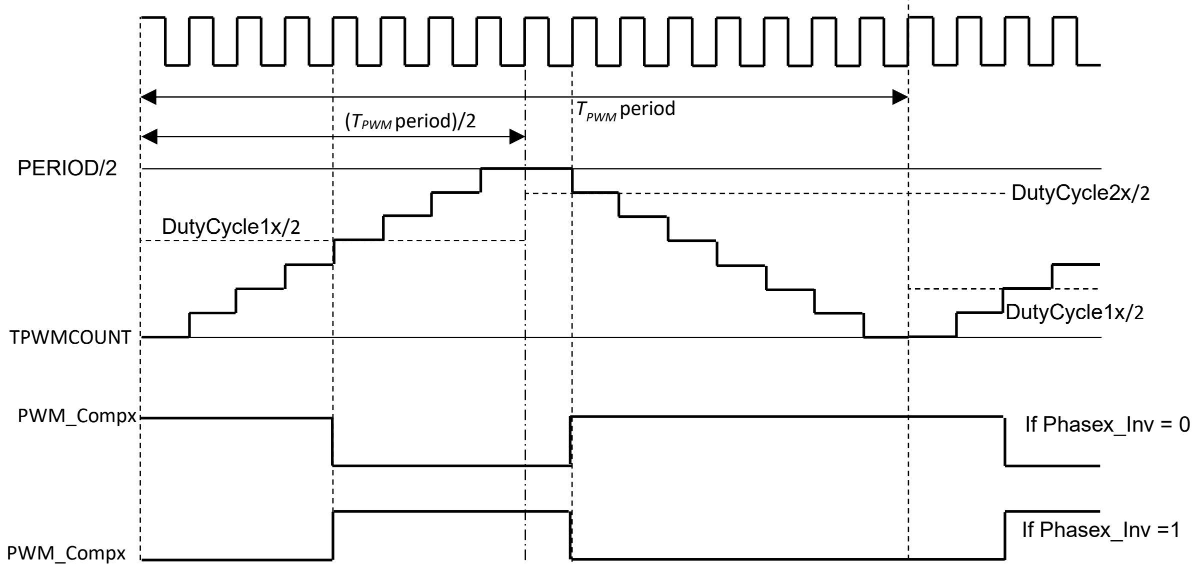

Figure 1-4: Compare Unit in Centre-Aligned Mode

1.5.2 Edge-Aligned Mode

Operation in edge-aligned mode depends on the value of DutyCycle1 and DutyCycle2 and falls into the following two conditions:

- **Condition 1**: DutyCycle1 < DutyCycle2
- **Condition 2**: DutyCycle1 ≥ DutyCycle2

Figure 1-5 shows the functional behaviour of the compare unit in the edge-aligned mode for the condition 1. The compare unit in edge-aligned mode (PWM_PWMMOD=0), compares the variable DutyCycle1x and DutyCycle2x to TPWMCOUNT to generate the output PWM_Compx to the dead-time control unit. When Phasex_Inv is set 1, PWM_Compx signal is inverted. The duty cycle of the PWM_Compx for the condition 1 is calculated according to below equations. It should be noted if DutyCycle1 or DutyCycle2 are set to a value greater than PERIOD it is automatically clamped to PERIOD in the hardware to avoid the overflow condition.

When Phasex_Inv = 0

$$PWM\_Compx\ (Duty\ Cycle) = \left(\frac{\text{DutyCycle1x}}{PERIOD} + \frac{PERIOD - \text{DutyCycle2x}}{PERIOD}\right) \times 100\%$$

When Phasex_Inv = 1

$$PWM\_Compx\ (Duty\ Cycle) = \left(1 - \left(\frac{\text{DutyCycle1x}}{PERIOD} + \frac{PERIOD - \text{DutyCycle2x}}{PERIOD}\right)\right) \times 100\%$$

Important Notes: The operations starts once the PERIOD is set to a value greater than 15_d.

: ‘x’ represents one of the three PWM generators

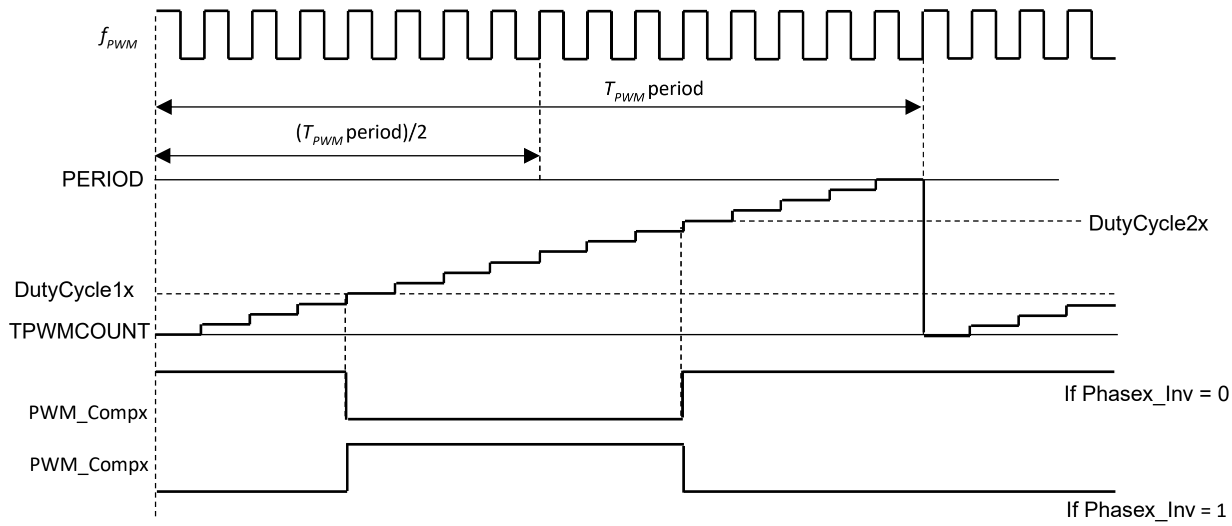

Figure 1-5: Edge-aligned mode operation for condition 1

Figure 1-6 shows the functional behaviour of the compare unit in the edge-aligned mode for the condition 2. In this mode, the compare unit in edge-aligned mode (PWM_PWMMOD=0), compares the variable DutyCycle1x to TPWMCOUNT to generate the output PWM_Compx to the dead-time control unit. In condition 2, DutyCycle2x is ignored and not used. When Phasex_Inv is set 1, PWM_Compx signal is inverted. The duty cycle of the PWM_Compx for the condition 2 is calculated according to below equations. It should be noted if DutyCycle1 is set to a value greater than PERIOD it is automatically clamped to PERIOD in the hardware to avoid the overflow condition.

When Phasex_Inv = 0

$$PWM\_Compx\ (Duty\ Cycle) = \left(\frac{\text{DutyCycle1x}}{PERIOD}\right) \times 100\%$$

When Phasex_Inv = 1

$$PWM\_Compx\ (Duty\ Cycle) = \left(1 - \frac{\text{DutyCycle1x}}{PERIOD}\right) \times 100\%$$

Important Notes: The operations starts once the PERIOD is set to a value greater than 15_d.
: 'x' represents one of the three PWM generators

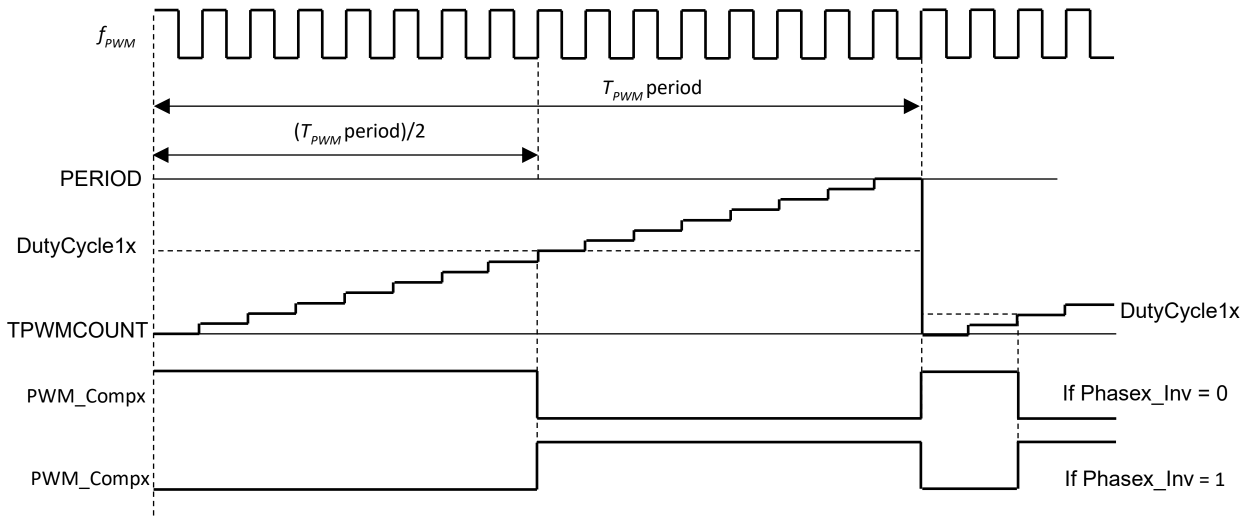

Figure 1-6: Edge-aligned mode operation for condition 2

1.6 Dead-Time Control Unit

Dead-time control unit processes PWM_Compx signal and generates complimentry signals, PWM_Hx and PWM_Lx, for the output stage control unit. The primary function of generating dead time is to avoid short circuit condition in the external MOSFETs in the power bridge circuit. When DeadTime in PWM dead time register is set to '0', PWM_Hx is set high and PWM_Lx is set low as soon as PWM_compx is set high and vice versa. When dead time is a none zero value, PWM_Hx and PWM_Lx are set high after DeadTime duration with respect to each other. The functional behaviour of dead-time control unit is shown in Figure 1-7. The dead time duration can be calculated according to the following formula:

$$Dead\ Time\ Duration = \frac{DeadTime}{f_{PWM}}\ \mathrm{s}$$

For example in order to achieve 1µs dead time duration, PWM_DT can be calculated as below for a f_{PWM} of 80MHz

$$DeadTime = 1\mu s \times 80MHz = 80$$

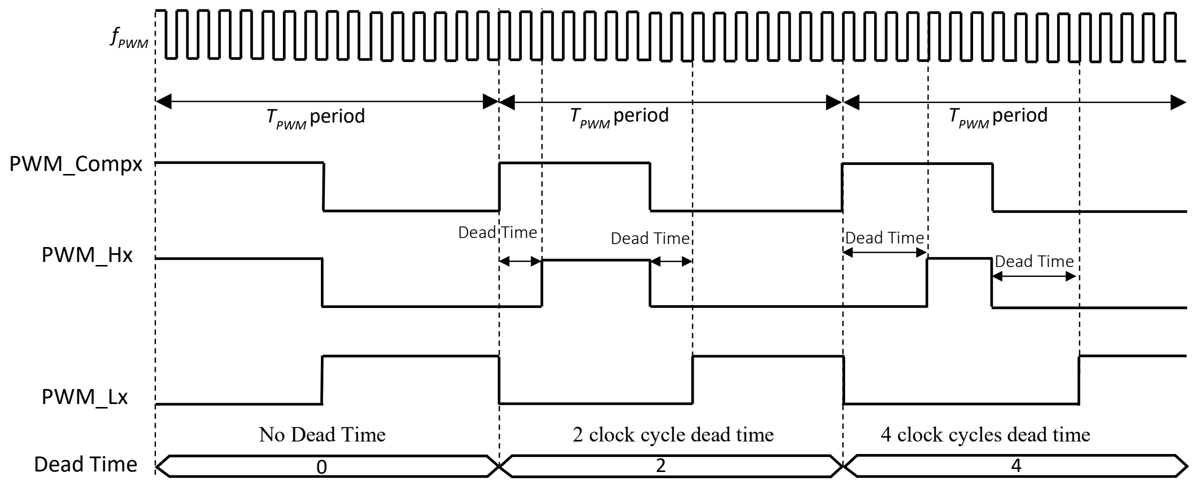

Figure 1-7: Dead Time Control Unit Functional Behavior

1.7 Output Stage Control Unit

The output stage control defines the final output Hx and Lx signals which are used to drive the external MOSFETs. The port connection for signals Hx and Lx needs to be defined by the GPIO module. Table 1-1 shows the truth logic table for signals HA and LA for phase A. The same truth logics are repeated for for the phases B and C.

Table 1-1: Truth Logics Table for Output Stage

PERIOD>15_d	OUTSEL_HA	PhaseDis_En	PhaseA_Dis	HA
X	0	X	X	LLVLSEL_HA
False	1	X	X	0
True	1	0	X	PWM_HA
True	1	1	0	PWM_HA
True	1	1	1	0

PERIOD>15_d	OUTSEL_LA	PhaseDis_En	PhaseA_Dis	LA
X	0	X	X	LLVLSEL_LA
False	1	X	X	0
True	1	0	X	PWM_LA
True	1	1	0	PWM_LA
True	1	1	1	0

X:don't care condition

1.8 Event Trigger Control Unit

This unit generates the trigger signals to the current acquisition unit (CAU) to sample the current in the power bridge circuit measured across the sense resistor. Two trigger signals, ADCANT and ADCACT are generated from this unit for phase A and similary 4 other trigger signals ADCBNT, ADCBCT, ADCCNT, ADCCCT are generated for Phases B and C respectively. ADC calibration triggers (ADCxCT) is connected to CAU module and is mainly used to calibrate the analog to digital converter. ADC normal trigger (ADCxNT) is mainly used to sample the current at a desired point in a PWM cycle. 'X' represent phase A, B or C. However, for applications which requires two trigger signal per each PWM cycle it is possible to use both of these trigger signals to sample the current measured across the sense resistor.

1.8.1 Analog to Digital Converter Normal Trigger (ADCxNT)

In this mode of operation ADCxNTREF defined by ADC normal trigger register is compared to Carrier signal to generate the trigger signal ADCxNT signal. The signal ADCxNT is connected to CAU module to sample the currnet measured across the sense resistor in the power bridge circuit. When Carrier value is greater than ADCxNTREF, ADCxNT is set high and it remains high untill 4 clock cycles before the end of T_{pwm} period. If ADCxNTREF is greater than (PERIOD-4), no trigger signal will be generated. This unit is enabled when PERIOD > 15_d and ADCxNT_EN is set to '1' in control2 register. Figure 1-8 demonstrates the functional behaviour of ADC normal trigger generation (ADCxNT). The trigger point for ADCxNTREF after the start of new T_{pwm} period can be calculated according to equation below:

$$Trigger\ Point\ for\ \text{ADCxNT} = \frac{\text{ADCxNTREF}}{f_{PWM}}\ \text{s}$$

For example if it is desired to generate the trigger signal ADCxNT after 2µs with respect to start of the new T_{pwm} period, ADCxNTREF can be calculated as below assuming 80MHz f_{PWM}.

$$\text{ADCxNTREF} = 80\text{MHz} \times 2\mu\text{s} = 160$$

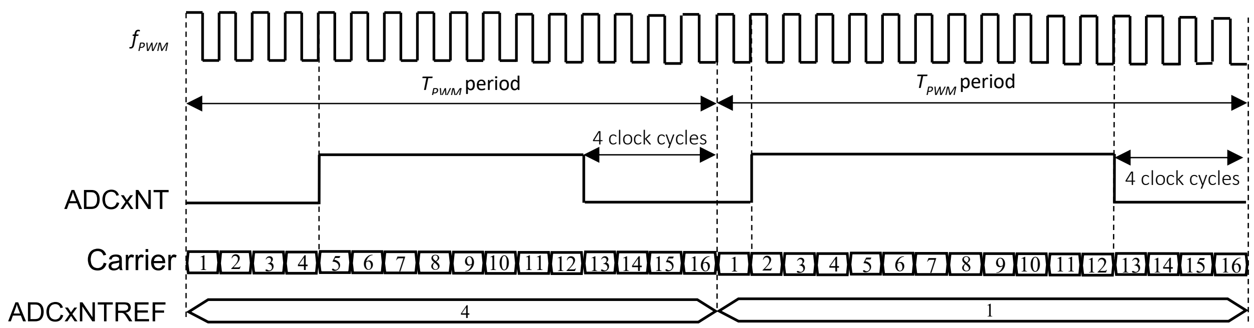

Figure 1-8: Analog to Digital Converter Normal Trigger Generation (ADCxNT)

1.8.2 Analog to Digital Converter Calibration Trigger (ADCxCT)

In this mode of operation, ADCxCTREF defined by ADC normal trigger register is compared to the Carrier signal to generate the trigger signal ADCxCT signal. When Carrier value is greater than ADCxCTREF, ADCxCT is set high and it remains high untill 4 clock cycles before the end of T_{pwm} period. If ADCxCTREF is greater than (PERIOD-4), no trigger signal will be generated. This unit is enabled when PERIOD > 15_d and ADCxNT_EN is set to '1' in control2 register. Figure 1-9 demonstrates the functional behaviour of ADC normal trigger generation (ADCxCT). The trigger point for ADCxCTREF after the start of new T_{pwm} period can be calculated according to the equation below:

$$\textit{Trigger Point for } \mathrm{ADCxCT} = \frac{\mathrm{ADCxCTREF}}{f_{PWM}} \ \mathrm{s}$$

For example if it is desired to generate the trigger signal ADCxCT after 2µs with respect to start of the new T_{pwm} period, ADCxCTREF can be calculated as below assuming 80MHz f_{PWM}.

$$\mathrm{ADCxCTREF} = 80\mathrm{MHz} \times 2\mu\mathrm{s} = 160$$

The signal ADCxCT is connected to CAU module to calibrate the analog to digital converter and also take an extra sample of current if desired. ADCxCT and ADCxNT allow sampling current twice per each phase in a PWM period, adding up to a total of 6 sampels if all three channels are used.

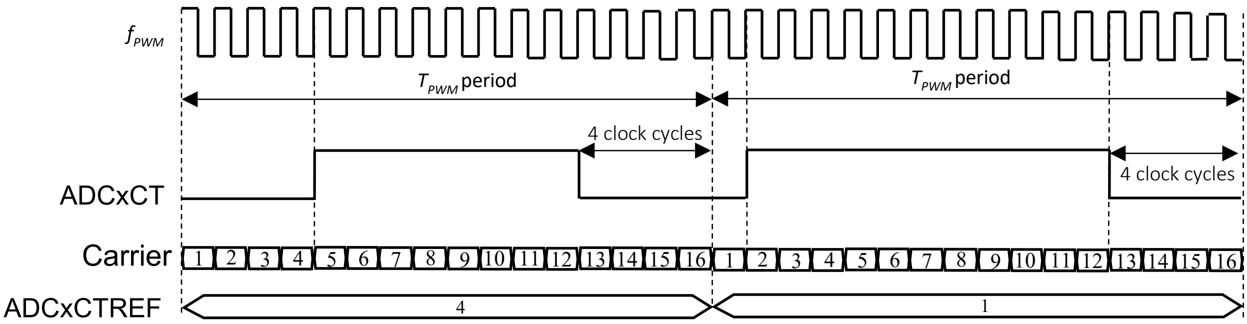

Figure 1-9: Analog to Digital Converter Calibration Trigger Generation (ADCxCT)

1.9 Register Update Synchronisation

It is possible to change the value of the PWM module registers at any time. However, the new value of the register for the operation is only effective at the start of the new PWM cycle. This is to ensure, the effect of all the new registers values are synchronsied with the start of PWM cycle. The synchronisation with PWM cycle is crucial to ensure correct PWM duty cycle, dead time and ADC trigger signal generation. If a value of a register is changed mutliple times before the start of a new PWM cycle, only the last change will be effective in the operation. Figure 1-10 illustrates the register update operation.

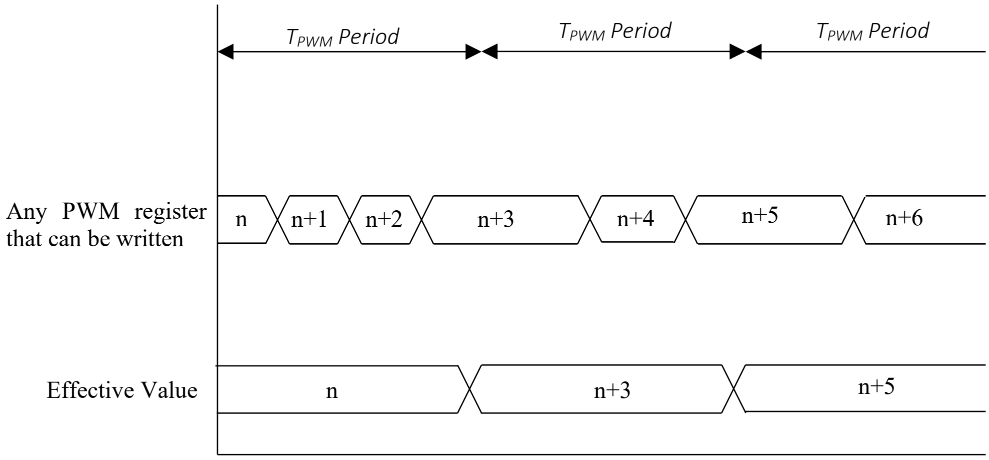

Figure 1-10: Register Update Synchronization

1.10 Register sets

The PWM module contains 17 registers in total. The register names, address offsets and access type and reset values are shown in table 1.

PWM Module Base Address: 0x40010000

Register Name	Address Offset	Access Type	Reset Value
PWM Duty Cycle Phase A	0x00	R/W	0x00000000
PWM Duty Cycle Phase B	0x04	R/W	0x00000000
PWM Duty Cycle Phase C	0x08	R/W	0x00000000
PWM Period Control	0x0C	R/W	0x00000000
Reserved	0x10	R/W	0x00000000
Reserved	0x14	R/W	0x00000000
Reserved	0x18	R/W	0x00000000
ADC Normal Trigger Reference Phase A	0x1C	R/W	0x00000000
ADC Normal Trigger Reference Phase B	0x20	R/W	0x00000000
ADC Normal Trigger Reference Phase C	0x24	R/W	0x00000000
ADC Calibration Trigger Reference Phase A	0x28	R/W	0x00000000
ADC Calibration Trigger Reference Phase B	0x2C	R/W	0x00000000
ADC Calibration Trigger Reference Phase C	0x30	R/W	0x00000000
Output Selection	0x34	R/W	0x00000000
Output Logic Level Selection	0x38	R/W	0x00000000
Control Register 1	0x3C	R/W	0x00000000
Control Register 2	0x40	R/W	0x00000000
PWM Dead Time	0x44	R/W	0x00000078
PWM Carrier	0x48	R	0x00000000
PWM Centre and Start	0x4C	R	0x00000000

PWM Duty Cycle

Reset Value: 0x00000000

Register Address Phase A: 0x40010000
Register Address Phase B: 0x40010004
Register Address Phase C: 0x40010008

This register defines the PWM output, PWM_Compx, duty cycle.

bit31	bit30	bit29	bit28	bit27	bit26	bit25	bit24
Res	Res	DutyCycle2[29:24]					
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bit23	bit22	bit21	bit20	bit19	bit18	bit17	bit16
DutyCycle2[23:16]							
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
Res	Res	DutyCycle1[13:8]					
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
DutyCycle1[7:0]							
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Legend:	
R: Read	U: Unimplemented bit, read as 0
W: Write	Res: Reserved bit, read as 0
1C: Write 1 to clear	

Field	Bits	Description
DutyCycle1	13:0	Defines the duty cycle of the PWM signal PWM_Compx depending on PWM mode of operation.
DutyCycle2	29:16	

Edge-Aligned Mode:

Condition 1: DutyCycle1 < DutyCycle2

If Phasex_Inv = 0

$$PWM\_Compx\ (DutyCycle) = \left(\frac{\text{DutyCycle1x}}{PERIOD} + \frac{PERIOD - \text{DutyCycle2x}}{PERIOD}\right) \times 100\%$$

If Phasex_Inv = 1

$$PWM\_Compx\ (DutyCycle) = \left(1 - \left(\frac{\text{DutyCycle1x}}{PERIOD} + \frac{PERIOD - \text{DutyCycle2x}}{PERIOD}\right)\right) \times 100\%$$

Condition 2: DutyCycle1 ≥ DutyCycle2

If Phasex_Inv = 0

$$PWM\_Compx\ (DutyCycle) = \left(\frac{\text{DutyCycle1x}}{PERIOD}\right) \times 100\%$$

If Phasex_Inv = 1

$$PWM\_Compx\ (DutyCycle) = \left(1 - \frac{\text{DutyCycle1x}}{PERIOD}\right) \times 100\%$$

Centre-Aligned Mode:

When Phasex_Inv = 0

$$PWM\_Compx(Duty\ Cycle) = \frac{\frac{\mathrm{DutyCycle1x}}{2} + \frac{\mathrm{DutyCycle1x}}{2}}{Period} \times 100\ \%$$

When Phasex_Inv = 1

$$PWM\_Compx\ (Duty\ Cycle) = \frac{\frac{(Period - \mathrm{DutyCycle1x})}{2} + \frac{(Period - \mathrm{DutyCycle2x})}{2}}{Period} \times 100\ \%$$

X: can reperesent phase A,B or C.

PWM Period Register Address : 0x4001000C
Reset Value: 0x00000000

This register defines the period for PWM output, PWM_Compx, period.

bit31	bit30	bit29	bit28	bit27	bit26	bit25	bit24
Res	Res	Res	Res	Res	Res	Res	Res
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bit23	bit22	bit21	bit20	bit19	bit18	bit17	bit16
Res	Res	Res	Res	Res	Res	Res	Res
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
Res	Res	PERIOD [13:8]					
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
PERIOD [7:0]							
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Legend:	
R: Read	U: Unimplemented bit, read as 0
W: Write	Res: Reserved bit, read as 0
1C: Write 1 to clear	

Field	Bits	Description
PERIOD	13:0	Defines the period of the generated PWM signal PWM_Compx.

$$\text{PWM\_Compx(period)} = \frac{\boldsymbol{PERIOD}}{f_{PWM}}\ s$$

Where f_{PWM} is the input clock frequency to PWM Module (80Mhz)

ADC Normal Trigger Reference
Reset Value: 0x00000000

Register Address Phase A: 0x4001001C
Register Address Phase B: 0x40010020
Register Address Phase C: 0x40010024

These registers define the normal trigger reference (ADCxNT) point generated to sample the current in every PWM cycle.

bit31	bit30	bit29	bit28	bit27	bit26	bit25	bit24
Res	Res	Res	Res	Res	Res	Res	Res
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bit23	bit22	bit21	bit20	bit19	bit18	bit17	bit16
Res	Res	Res	Res	Res	Res	Res	Res
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
Res	Res	ADCNTREF [13:8]					
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
ADCNTREF [7:0]							
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Legend:	
R: Read	U: Unimplemented bit, read as 0
W: Write	Res: Reserved bit, read as 0
1C: Write 1 to clear	

Field	Bits	Description
ADCxNTREF	13:0	This parameters defines the point at which ADCxNT is set high and sent to CAU module to sample the current across the sense resistor. The point at which this trigger signal is set high with respect to start of the every PWM cycle can be calculated as below: $$ADCNT = \frac{\mathbf{ADCxNTREF}}{f_{PWM}}\ s$$ If **ADCxNTREF** is set to a value greater than (**PERIOD** – 4), ADCNT is never set high for that particular T_{PWM} period. X: can reperesent phase A,B or C.

ADC Calibration Trigger Reference
Reset Value: 0x00000000

Register Address Phase A: 0x40010028
Register Address Phase B: 0x4001002C
Register Address Phase C: 0x40010030

These registers define the calibration trigger reference point (ADCxCT) generated to sample the current in every PWM cycle. The trigger signals generated by programming these registers are also used for calibration of analog to digital converter used in CAU module to sample the current across the shunt resistor.

bit31	bit30	bit29	bit28	bit27	bit26	bit25	bit24
Res	Res	Res	Res	Res	Res	Res	Res
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bit23	bit22	bit21	bit20	bit19	bit18	bit17	bit16
Res	Res	Res	Res	Res	Res	Res	Res
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
Res	Res	ADCCTREF [13:8]					
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
ADCCTREF [7:0]							
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Legend:	
R: Read	U: Unimplemented bit, read as 0
W: Write	Res: Reserved bit, read as 0
1C: Write 1 to clear	

Field	Bits	Description
ADCxCTREF	13:0	This parameters defines the point at which ADCxCT is set high and sent to CAU module to sample the current across the sense resistor. The point at which this trigger signal is set high with respect to start of the every PWM cycle can be calculated as below: $$ADCNT = \frac{\textbf{ADCxCTREF}}{f_{PWM}}\ s$$ ADCxCT generated by programming this register is also used to calibrate the ADC used to sample the current across the shunt resistor. If **ADCxCTREF** is set to a value greater than (**PERIOD** – 4), ADCCT is never set high for that particular T_{PWM} period. X: can reperesent phase A,B or C.

Output Selection Register Address Phase A: 0x40010034
Reset Value: 0x00000000

This register select the bits for the multiplexer that defines if the output signals, Hx and Lx, should be generated from independent PWM generator or by **output level selection** register. This register also defines if the physical PhasexDis signals connected from AMCT module to PWM module should be ignored or not when the output is decided by PWM generator.

bit31	bit30	bit29	bit28	bit27	bit26	bit25	bit24
Res	Res	Res	Res	Res	Res	Res	Res
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bit23	bit22	bit21	bit20	bit19	bit18	bit17	bit16
Res	Res	Res	Res	Res	Res	Res	Res
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
Res	Res	Res	Res	Res	Res	Res	Res
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
Res	PhaseDis_En	OUTSEL_HC	OUTSEL_LC	OUTSEL_HB	OUTSEL_LB	OUTSEL_HA	OUTSEL_LA
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Legend:
R: Read
W: Write
1C: Write 1 to clear
U: Unimplemented bit, read as 0
Res: Reserved bit, read as 0

Field	Bits	Description
OUTSEL_Hx	5,3,1	When OUTSEL_Hx is set to 0, the output Hx signal is defined by LLVLSEL_Hx value defined by output logic level selection register. Otherwise the output is decided by independent PWM generators. Table below summerizes the output in different conditions.

PERIOD>15_d	OUTSEL_Hx	PhaseDis_En	PhaseA_Dis	Hx
Don't care	0	X	X	LLVLSEL_Hx
False	1	X	X	0
True	1	0	X	PWM_Hx
True	1	1	0	PWM_Hx
True	1	1	1	0

Field	Bits	Description
OUTSEL_Lx	4,2,0	When OUTSEL_Lx is set to 0, the output Lx signal is defined by LLVLSEL_Lx value defined by output logic level selection register. Otherwise the output is decided by independent PWM generators. Table below summerizes the output in different conditions.

PERIOD>15_d	OUTSEL_Lx	PhaseDis_En	PhaseA_Dis	Lx
Don't care	0	X	X	LLVLSEL_Lx
False	1	X	X	0
True	1	0	X	PWM_Lx
True	1	1	0	PWM_Lx
True	1	1	1	0

X can represent Phase A, B or C.

Output Logic Level Selection Register Address: 0x40010038
Reset Value: 0x00000000

This register defines the logic level output for signals Hx and Lx when the corresponding OUTSEL_Hx and OUTSEL_Lx is set to ‘0’ in **output selection** register.

bit31	bit30	bit29	bit28	bit27	bit26	bit25	bit24
Res	Res	Res	Res	Res	Res	Res	Res
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bit23	bit22	bit21	bit20	bit19	bit18	bit17	bit16
Res	Res	Res	Res	Res	Res	Res	Res
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
Res	Res	Res	Res	Res	Res	Res	Res
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
Res	Res	LLVLSEL_HC	LLVLSEL_LC	LLVLSEL_HB	LLVLSEL_LB	LLVLSEL_HA	LLVLSEL_LA
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Legend:
R: Read
W: Write
1C: Write 1 to clear
U: Unimplemented bit, read as 0
Res: Reserved bit, read as 0

Field	Bits	Description
LLVLSEL_Hx	5,3,1	The signal Hx = LLVLSEL_Hx when OUTSEL_Hx is set to 0.
LLVLSEL_Lx	4,2,0	The signal Lx = LLVLSEL_Lx when OUTSEL_Lx is set to 0.

X can represent Phase A, B or C.

Control Register 1 Register Address: 0x4001003C
Reset Value: 0x00000000

This register defines the PWM output mode. This register also defines if the PWM output ,PWM_Compx, should be inversed or not in the desired mode.

bit31	bit30	bit29	bit28	bit27	bit26	bit25	bit24
Res	Res	Res	Res	Res	Res	Res	Res
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bit23	bit22	bit21	bit20	bit19	bit18	bit17	bit16
Res	Res	Res	Res	Res	Res	Res	Res
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
Res	Res	Res	Res	Res	Res	Res	Res
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
Res	Res	Res	PhaseC_Inv	PhaseB_Inv	PhaseA_Inv	Res	PWMMOD
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Legend:	
R: Read	U: Unimplemented bit, read as 0
W: Write	Res: Reserved bit, read as 0
1C: Write 1 to clear	

Field	Bits	Description
Phasex_Inv	4,3,2	0: No change in the signal PWM_Compx 1: Invert the signal PWM_Compx
PWMMOD	0	0: Edge-Aligned System 1: Centre-Aligned System

X can represent Phase A, B or C.

Control Register 2

Register Address: 0x40010040

Reset Value: 0x00000000

The parameters in this register enable or disable ADC normal trigger and ADC calibration trigger signals. When disabled, no trigger signals will be generated.

bit31	bit30	bit29	bit28	bit27	bit26	bit25	bit24
Res	Res	Res	Res	Res	Res	Res	Res
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bit23	bit22	bit21	bit20	bit19	bit18	bit17	bit16
Res	Res	Res	Res	Res	Res	Res	Res
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
Res	Res	Res	Res	Res	Res	Res	ADCCCT_EN
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
ADCBCT_EN	ADCACT_EN	ADCCNT_EN	ADCBNT_EN	ADCANT_EN	Res	Res	Res
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Legend:
R: Read
W: Write
1C: Write 1 to clear
U: Unimplemented bit, read as 0
Res: Reserved bit, read as 0

Field	Bits	Description
ADCxCT_En	8,7,6	0: Disable the ADC calibration trigger signal ADCxCT. 1: Enable the ADC calibration trigger signal ADCxCT.
ADCxNT_En	5,4,3	0: Disable the ADC normal trigger signal ADCxNT. 1: Enable the ADC normal trigger signal ADCxNT.

X can represent Phase A, B or C.

PWM Dead Time
Register Address: 0x40010044
Reset Value: 0x00000078

This register defines the deadtime between the complementary signals PWM_Hx and PWM_Lx. The deadtime is only generated in PWM generation mode (OUTSELHx = 1 for PWM_Hx and OUTSELx=1 for PWM_Lx).

bit31	bit30	bit29	bit28	bit27	bit26	bit25	bit24
Res	Res	Res	Res	Res	Res	Res	Res
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bit23	bit22	bit21	bit20	bit19	bit18	bit17	bit16
Res	Res	Res	Res	Res	Res	Res	Res
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
Res	Res	Res	Res	Res	Res	DeadTime[9:8]	
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
DeadTime[7:0]							
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Legend:	
R: Read	U: Unimplemented bit, read as 0
W: Write	Res: Reserved bit, read as 0
1C: Write 1 to clear	

Field	Bits	Description
DeadTime	9:0	Defines the dead time duration for the complementary signals PWM_Hx and PWM_Lx. The dead time duration can be caluclated based on the following eqaution: $\text{Dead Time Duration} = \frac{\boldsymbol{DeadTime}}{f_{PWM}}\ s$ The deadtime is only generated in PWM generation mode (OUTSELHx = 1 for PWM_Hx and OUTSELx=1 for PWM_Lx).

Carrier information Register Address: 0x40010048
Reset Value: 0x00000000

This registers contains the count value of the carrier signal which counts from 1 to PERIOD (defined by **PWM period** register).

bit31	bit30	bit29	bit28	bit27	bit26	bit25	bit24
Res	Res	Res	Res	Res	Res	Res	Res
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bit23	bit22	bit21	bit20	bit19	bit18	bit17	bit16
Res	Res	Res	Res	Res	Res	Res	Res
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
Res	Res	CARRIER[13:8]					
R/W	R/W	R	R	R	R	R	R

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
CARRIER[7:0]							
R	R	R	R	R	R	R	R

Legend:
R: Read
W: Write
1C: Write 1 to clear
U: Unimplemented bit, read as 0
Res: Reserved bit, read as 0

Field	Bits	Description
CARRIER	13:0	Contains the count value for the carrier signal that counts up to PERIOD and resets to 1 once it reaches to PERIOD value defined by PWM period register.

PWM Centre and Start Register Address: 0x4001004C
Reset Value: 0x00000000

This register can be read to understand the start and centre point of T_{pwm} period.

bit31	bit30	bit29	bit28	bit27	bit26	bit25	bit24
Res	Res	Res	Res	Res	Res	Res	Res
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bit23	bit22	bit21	bit20	bit19	bit18	bit17	bit16
Res	Res	Res	Res	Res	Res	Res	Res
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
Res	Res	Res	Res	Res	Res	Res	Res
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
Res	Res	Res	Res	Res	Res	Res	PWM_Centre
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R

Legend:	
R: Read	U: Unimplemented bit, read as 0
W: Write	Res: Reserved bit, read as 0
1C: Write 1 to clear	

Field	Bits	Description
PWM_Centre	0	PWM_Centre indicates the start and centre point of T_{pwm} period. PWM_Centre is set low at the start of T_{pwm} period and it is set high at the centre of T_{pwm} period.

Revision Control:

Feb/22/2021	Initial Version
Nov/11/2021	• Modifed Edge-aligned section to reflect the latest changes • Updated Logo